

BAUST - Department of Computer Science & Engineering

CSE 2104: Database Systems Sessional | Lab Manual 04 - Complex Queries & Joins

Objective: Master Nested Subqueries, Correlated Subqueries, and Outer Joins in Oracle SQL.

1. Experimental Schema

- EMP(emp_id PK, emp_name, dept_id FK, salary, hire_date, manager_id FK)
- DEPT(dept_id PK, dept_name, location)

2. Laboratory Tasks

Task 1: Retrieve employees who earn higher than the average salary of their respective department.

```
SELECT emp_name, salary, dept_id FROM EMP e WHERE salary > (  
    SELECT AVG(salary) FROM EMP WHERE dept_id = e.dept_id  
);
```

Task 2: Demonstrate difference between LEFT OUTER JOIN and FULL OUTER JOIN with NULL handling.
BAUST CSE Sessional Manual - Peer's Charity - Batch 11 Lab Repository

Task 3: Execute query execution plan using EXPLAIN PLAN FOR and identify Full Table Scan.